SMARTBRIDGE

SKILL WALLET
A SmartBridge Product

Code-Layout, Readability and Reusability

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Code Quality Checklist

#	Quality Criterion	Status	Evidence
1	**Consistent Indentation**	☑ Pass	All Python files use 4-space indentation; HTML templates use 2-space indentation consistently
2	**Meaningful File Names**	☑ Pass	train_model.py, app.py, preprocess_input(), build_target(), engineer_features() — all self-descriptive
3	**Meaningful Variable Names**	☑ Pass	BAD_STATUSES, CATEGORICAL_COLS, NUMERIC_COLS, scale_pos_weight, IS_UNEMPLOYED — no ambiguous single-letter names
4	**Modular Function Design**	☑ Pass	train_model.py split into: load_data(), build_target(), engineer_features(), preprocess(), train_all_models(), save_plots()
5	**Single Responsibility**	☑ Pass	Each function handles one task; app.py handles routing only, preprocessing is isolated in preprocess_input()
6	**Reusable Preprocessing**	☑ Pass	encoder.pkl and scaler.pkl are shared between training (train_model.py) and inference (app.py) — single source of truth
7	**No Redundant Code**	☑ Pass	Model training loop avoids copy-paste; model comparison is data-driven via the models dict
8	**Error Handling**	☑ Pass	MODELS_LOADED flag in app.py prevents crashes when PKL files are missing; Flask handles invalid routes gracefully

Overall Code Quality Rating: 5 / 5

Project File Structure

```
Credit Card Approval Prediction/
├── app.py                    # Flask web application (4 routes)
├── train_model.py            # ML training pipeline
├── requirements.txt          # Python dependencies
├── Procfile                  # Gunicorn start command for Render
├── render.yaml               # Render deployment configuration
├── .gitignore                # Excludes data/, .env, __pycache__
│
├── templates/                # Jinja2 HTML templates
│   ├── base.html             # Shared layout: navbar, Bootstrap CDN
│   ├── index.html            # Dashboard: model stats + scenario cards
│   ├── predict.html          # Prediction form + result display
│   └── history.html          # Session history listing
│
├── static/
│   ├── css/style.css         # Banking theme: navy + gold CSS variables
│   └── js/script.js          # Scenario pre-fill + confidence bar JS
│
├── models/                   # Trained model artifacts (committed to git)
│   ├── model.pkl             # Best model: Random Forest
│   ├── scaler.pkl            # StandardScaler (fitted on training data)
│   ├── encoder.pkl           # LabelEncoders (5 categorical features)
│   └── metadata.pkl          # Model comparison dict + feature column order
│
├── notebooks/
│   └── Credit_Card_Approval_Analysis.ipynb  # EDA + model experimentation
│
└── docs/                     # 8-phase project documentation (this folder)
```